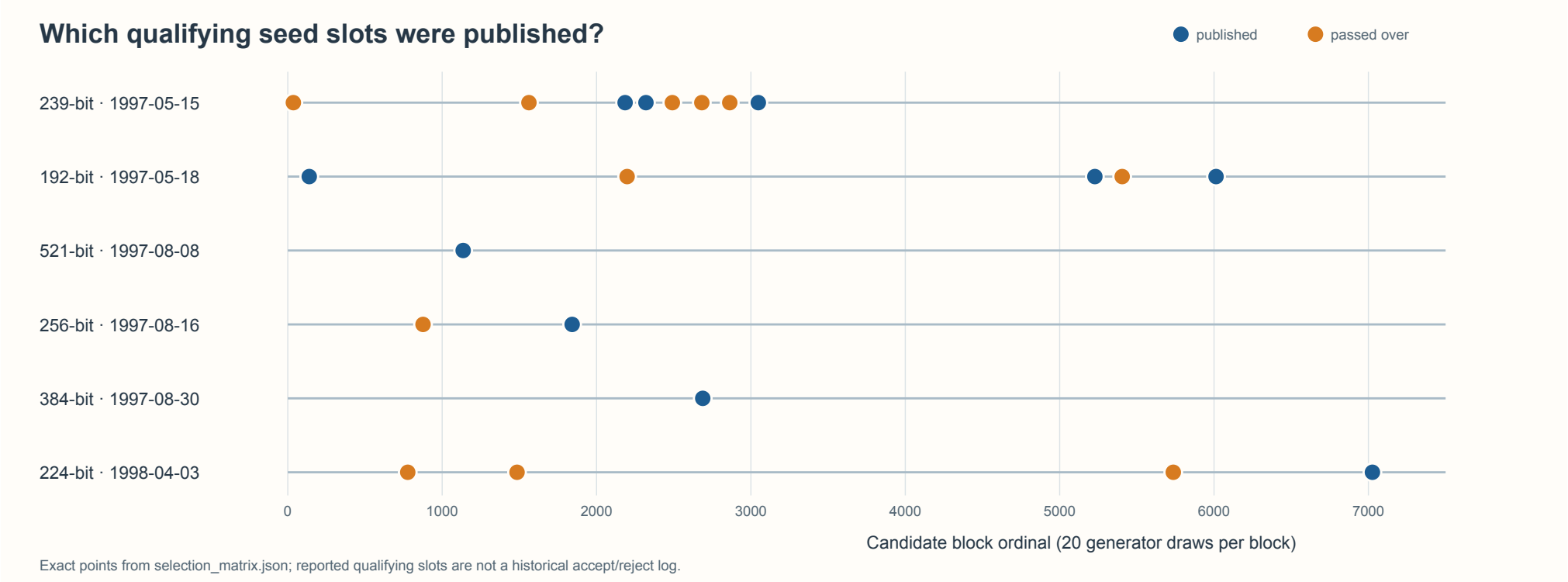
Why did the eleven qualifying seed candidates go unpublished?

Historical selection audit · 16 October 2026 · exact data and code in the accompanying study directory



Supported answer

The recovered timestamp-initialized lrand48 streams reproduce all ten published prime-field seeds. Eleven other reported 20-byte blocks in those streams produce valid prime-order curves under the same seed-to-coefficient procedure. Four precede a NIST curve: one before P-256 and three before P-224. Seven lie in the two May 1997 streams that produced the ANSI example curves. A qualifying block in a reconstructed stream is not proof that the original program evaluated and rejected it.

The checked ordinary criteria do not separate the published curves from the passed-over ones. Rules requiring $n < p$, positive trace, prime twist, no small twist factors, or the first qualifying block all fail a published control. A bounded PARI/GP factorization comparison also refutes a uniform preference for a larger prime factor of the twist order: some passed-over 192- and 239-bit curves have larger factors than an earlier published curve over the same field. A historical cause remains unknown.

What the historical standard establishes

FIPS 186-2 Appendix 6.4 says to choose an arbitrary 160-bit seed, derive the curve coefficient through SHA-1, and test for suitable order. It does not prescribe the seed progression or retain an acceptance log. Appendix 6.5 lets a reader verify the published seed-to-coefficient relation only. The source-bound earlier assessment checked the eleven exact groups by Sage point counting, proven-prime order, point-order and Hasse uniqueness, and found cofactor one for all. Its 21-curve bounded modular screen found no embedding degree at or below one million; the skipped classes had no CM-field discriminant at or below two million in absolute value. Those bounds are not complete endomorphism-ring or isogeny audits.

Selection hypotheses and counterexamples

Candidate rule	Observed counterexample or missing evidence	Result
First qualifying block	P-256 follows slot 876; P-224 follows slots 777, 1485 and 5736. The 239-bit stream is interleaved.	Refuted
Require n < p or positive trace	Published prime239v2 has n > p and negative trace. Eight skipped classes have n < p.	Refuted globally
Require no small twist factor	Nine of ten published curves have a twist factor at most 10,000.	Refuted globally
Require a larger twist prime factor	Skipped 192-bit slots 2198 and 5405 have largest certified factors of 170 and 181 bits; published prime192v1 has 95 bits.	Refuted globally
Per-field twist or other rule	P-224 has 118-bit largest twist factor versus 83 and 106 bits for two omissions, but slot 1485 timed out. No criterion or historical record was found.	Open
Same RNG consumed by other work	All published seeds remain aligned to 20-draw boundaries; May 1997 published base points use a distinct FreeLIP generator. Other calls remain possible.	Open
Historical arithmetic or bookkeeping error	Current computations confirm eleven suitable orders, but no historical program or defect is available to replay.	Open

The eleven passed-over classes

The ordinal is a one-based 20-byte block position. Trace is $t/(2\sqrt{p})$, where $t = p + 1 - n$. L is the bit length of the largest certified prime factor of the quadratic-twist order after PARI/GP factorization. A question mark means the 20-second residual-factorization limit expired. Full seed bytes, curve equations, group orders and EC1 identities are in the linked earlier assessment.

Field / slot	Trace	n > p	Twist factors ≤ 10,000	L	Relevant comparison
239 / 36	+0.730	No	139	138	Published prime239v1 has L = 86.
239 / 1562	−0.113	Yes	41	179	Published prime239v2 also has n > p.
239 / 2491	−0.151	Yes	43, 173, 307, 953	129	Small twist factors occur in published controls.
239 / 2682	−0.062	Yes	29	?	Square factor of 5 in Frobenius discriminant also occurs in prime239v2.
239 / 2863	+0.447	No	none found	125	No twist factor ≤ 10,000, yet passed over.
192 / 2198	+0.815	No	11, 127, 5827	170	Remaining twist factor is prime; published P-192 has L = 95.
192 / 5405	+0.472	No	7, 19^2	181	Remaining twist factor is prime; other published 192-bit L values are 139 and 161.
256 / 876	+0.156	No	3, 11, 37, 8081	201	P-256 has L = 241; one pair does not establish a rule.
224 / 777	+0.376	No	3, 5, 683	83	P-224 has L = 118.
224 / 1485	+0.015	No	3, 31	?	Complete twist factorization timed out.
224 / 5736	+0.311	No	3, 1607, 3371	106	P-224 has L = 118.

Inference boundary

The 18 completed twist factorizations give exact factors; three residuals timed out, so their L is unknown. A large twist prime factor is a structural feature, not an ECDLP or protocol-security verdict. The two P-224 coefficient signs for each seed are isomorphic, leaving three seed-level classes. Every ordinary group order here is prime, so factors of n − 1 are irrelevant to Pohlig–Hellman on that group.

The reported stream slots do not reveal how many timestamps or alternative generator runs were tried. A 30-bit search-space illustration assumes a year-scale time window and a small fixed candidate cap; it is not a measured historical upper bound. Deliberate selection of a weak instance, an unpublished method applicable to honest instances, and an implementation flaw require separate evidence. An explicit hidden route would also need subgroup-preserving maps, recovery and verification, discovery cost, and cold-start versus amortized cost accounting.

The unresolved historical step is tracked in problems/nist-seed-selection-rule-unexplained.md. A provenanced original generator and call trace, or a predeclared rule that predicts independent historical decisions, would be discriminating evidence. The new stream plot changes no established isogeny or attack edge in knowledge/frontiers/ecdlp/.

Sources and reproducibility

Retrieved: Anthropic's discovery, Valsorda's candidate list, and FIPS 186-2 Appendices 6.4–6.5. Internal exact inputs: assessed.json, traits.json, supplemental_checks.json. New reproducible data: selection_matrix.json and twist_factor_probe.json, generated by the adjacent scripts. Full citations and limits are in report.md.